Practical No. 4

Aim: Write a java program demonstrating constructor overloading program.

Code:

Program 1:

```
class Rectangle {
    private double length;
    private double width;

    // 1. Default Constructor (No parameters - Unit Square)
    public Rectangle() {
        this.length = 1.0;
        this.width = 1.0;
    }

    // 2. Overloaded Constructor (One parameter - Square)
    public Rectangle(double side) {
        this.length = side;
        this.width = side;
    }

    // 3. Overloaded Constructor (Two parameters - Rectangle)
    public Rectangle(double length, double width) {
        this.length = length;
        this.width = width;
    }

    public double calculateArea() {
        return length * width;
    }

    public void displayDetails() {
        System.out.println("Length: " + length + ", Width: " + width + " | Area: " + calculateArea());
    }
}

public class Program1 {
    public static void main(String[] args) {
        System.out.println("=== Rectangle Constructor Overloading ===");

        // Calls Default Constructor
        Rectangle rect1 = new Rectangle();
        rect1.displayDetails();

        // Calls Single-Parameter Constructor
        Rectangle rect2 = new Rectangle(5.0);
        rect2.displayDetails();

        // Calls Two-Parameter Constructor
        Rectangle rect3 = new Rectangle(4.0, 6.0);
        rect3.displayDetails();
    }
```

```
}
```

Program 2:

```
class Book {
    private String title;
    private String author;
    private double price;

    // 1. Constructor with 1 parameter
    public Book(String title) {
        this(title, "Unknown Author", 0.0); // Chains to Constructor 3
    }

    // 2. Constructor with 2 parameters
    public Book(String title, String author) {
        this(title, author, 0.0); // Chains to Constructor 3
    }

    // 3. Fully Parametrized Constructor with 3 parameters
    public Book(String title, String author, double price) {
        this.title = title;
        this.author = author;
        this.price = price;
    }

    public void printSummary() {
        System.out.println("Title: \"" + title + "\" | Author: " + author + " | Price: $" + price);
    }
}

public class Program2 {
    public static void main(String[] args) {
        System.out.println("=== Book Constructor Overloading ===");

        // Creating objects using different constructors
        Book book1 = new Book("Java Basics");
        Book book2 = new Book("Effective Java", "Joshua Bloch");
        Book book3 = new Book("Clean Code", "Robert C. Martin", 45.99);

        book1.printSummary();
        book2.printSummary();
        book3.printSummary();
    }
}
```

Output:

```
=== Book Constructor Overloading ===
Title: "Java Basics" | Author: Unknown Author | Price: $0.0
Title: "Effective Java" | Author: Joshua Bloch | Price: $0.0
Title: "Clean Code" | Author: Robert C. Martin | Price: $45.99
```

```
=== Rectangle Constructor Overloading ===
Length: 1.0, Width: 1.0 | Area: 1.0
Length: 5.0, Width: 5.0 | Area: 25.0
Length: 4.0, Width: 6.0 | Area: 24.0
```